SK60B

USER MANUAL

DCS SK60

USER MANUAL

CONTENTS

INTRODUCTION

Welcome to the **SK60** for DCS World.

The Saab 105—known in Swedish Air Force service as the SK60—is a lightweight, twin-engine jet trainer renowned for its responsive handling, excellent visibility and remarkable versatility. For decades, it served as the foundation of Swedish military flight training while also fulfilling liaison, light-attack and aerobatic roles. The version we've chosen to create is the later version of the SK60 with the RM15 (RR-Williams FJ44) engines but without the most recent avionics upgrade.

This mod aims to recreate the distinctive experience of flying the SK60 in DCS World. Its approachable handling makes it well suited to basic flight training and relaxed sightseeing, while its navigation systems, weapon capabilities and multicrew cockpit offer plenty of depth for more advanced missions. The aircraft is equally at home flying low-level navigation routes, practicing formation maneuvers or conducting light ground-attack sorties.

This manual introduces the aircraft's cockpit, systems, normal procedures and operating techniques in its current state. It is intended both for pilots encountering the SK60 for the first time and for experienced DCS players who want to understand the aircraft in greater detail.

Take your time, learn the aircraft step by step—and above all, enjoy flying one of Sweden's most distinctive aircraft.

It's been a diverse team of people that's been working on this project over the years (we started back in 2020) and, even though there are always going to be areas to improve upon, we're really proud to share it with you. Hope you'll enjoy it as much as we do!

Cheers!

//Breadmaker

COCKPIT OVERVIEW

The SK60 is primarily a twin-seater trainer jet. What makes it stand out compared to many other trainers is the fact that pilot and instructor are sitting side-by-side instead of tandem. That gives both of them a very good visual understanding of what's happening since they are both seeing the same thing at the same time. It's also very easy for the instructor to give hands on instructions and help the student understand what needs to be done.

Our cockpit model has been carefully modelled based on reference photos, 3D scans and blueprints to give an authentic representation of the real thing. Most things are functional and work as they should. Here's an overview of the part of the cockpit you'll interact most with – the front panel.

1. Speed indicator
2. G-indicator
3. Airbrake indicator
4. Oxygen panel
5. Fuel Gauge
6. HSI
7. Variometer
8. Turn & Bank indicator
9. Main control panel
10. Flaps lever
11. Taxi/Landing lights switch
12. EADI
13. Altimeter
14. Backup ADI
15. N1 RPM indicators
16. RMI
17. FR31 Radio (UHF)
18. N2 RPM indicators
19. NAV units (VOR/DME, RNAV, ILS & ADF)
20. Audio panel (inop)
21. Cockpit lighting & Nav control panel
22. Exterior lighting panel
23. Gyro compass
24. Oil temp indicator
25. ITT indicators
26. Oil pressure indicator
27. Warning lights panel

Main Control Panel

Here’s what you’ll find on the main control panel:

1. **HUVUDSTROM** – Main Electrical Power
    a. **FPL-NÄT** – Main Electrical Power switch
    b. **LAMPTABLÅ** - inop
    c. **OMF I** – Inverter 1
    d. **OMF II** – Inverter 2
2. **MOTORSTART** – Engine Start
    a. Left and right LP fuel valve (lower)
    b. Left and right starter switches (upper)
3. **BRANSLE/PROV** – Fuelmanagement and testing switches (currently inop)
4. **LANDSTALL** – Landing Gear Lever
5. **GLÖM EJ LANDST O KLAFF/LANDST. LÅST UTE** – Landing gear and flap configuration warning
6. **BEVÄPNING** – Weapons arming panel (set to TILL for armed)

Interior Lighting Panel

1. INSTR LYSE – Cockpit Backlighting
2. INNERLYSE – Cockpit flood lighting
3. NAV – Navigation instruments power switch

Exterior Lighting panel

1. **LANTERN** – Navigation lights (Off/Bright/Blink/Dim)
2. **KOLL VARN** – Anti-collision lights (Off/Bright/Dim)

GETTING AIRBORNE – AND HOME AGAIN

Cold start procedures

This is how you start the SK60 mod. For full realistic procedures, see the checklist included in the kneeboard.

1. **FPL-NÄT** – TILL
2. **OMF I** – TILL
3. Set **interior lights** as desired.
4. **KOLL VARN** – HEL
5. Close **canopy**
6. Left **HP switch** – TILL
7. Left **engine starter** switch – TILL
8. Monitor left **N2 RPM**. When reaching 12% put left engine throttle lever in IDLE
9. Monitor left **N2 RPM** until it settles.
10. **Repeat** steps 4-7 for right engine.
11. **OMF II** – AUTO
12. **NAV** – TILL
13. Configure **NAV unit** as required
14. Set **exterior lights** as required
15. Set **taxi lights** as required
16. Check **main warning panel** to make sure all lights are out

Takeoff

1. Lineup on the runway.
2. Apply wheel brakes.
3. Set flaps to starting position (S)
4. Switch on landing lights.
5. Apply full throttle.
6. Release wheel brakes.
7. Use NWS until airspeed is alive.
8. Rotate around 200km/h.
9. Retract landing gear when positive climb rate is achieved.
10. Retract flaps.
11. Check gear & flaps indicator lights and verify they're off.
12. Set trim as desired and have a nice flight.

Landing

1. Decelerate below 400km/h.
2. Extend landing gear & flaps to "S" position.
3. Continue decelerating. Aim for 300km/h at base turn. Use airbrakes if necessary.
4. Lineup for final and reduce speed to 250km/h.
5. When lined up on short final and confident you're going to land, set flaps to "L" (landing).
6. Reduce speed to reach 210km/h over the threshold.
7. Set throttles to idle when over threshold and flare carefully to touch down with your main landing gears while keeping the nose up.
8. Keep holding the nose up for aerodynamic breaking.
9. Carefully apply wheel brakes (they lock and induce sliding if used to aggressively) and use NWS to turn onto an appropriate taxiway once the speed is low enough.
10. Retract flaps and airbrake if needed.

NAVIGATION SYSTEM

Overview

The SK60 version we've modelled uses a navigation system called the **KNS-81**.

The KNS-81 is an area-navigation system used for VOR/DME navigation. In addition to navigating directly to a conventional VOR/DME station, it allows the crew to define a waypoint at a selected bearing and distance from that station. This makes it possible to navigate to locations that are not marked by a physical radio beacon.

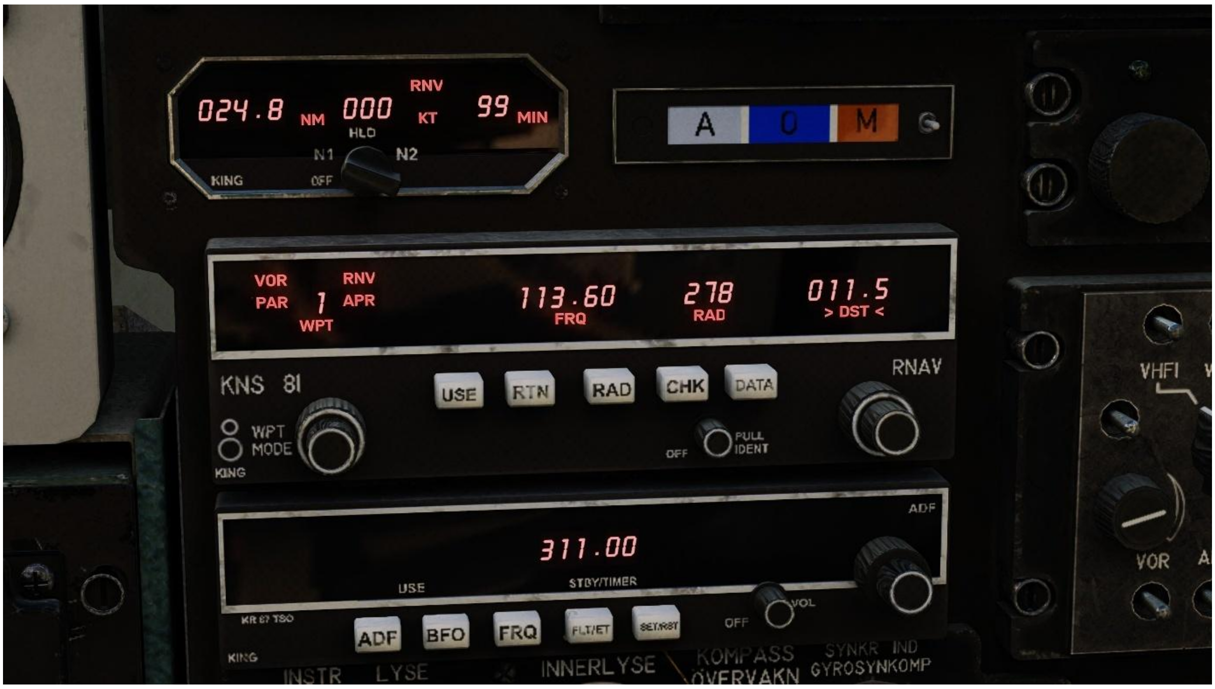
To define a waypoint, the crew selects a suitable VOR/DME station and enters the desired radial and distance. The KNS-81 then calculates the aircraft's position relative to the resulting waypoint and provides navigation guidance through the cockpit instruments. The system can also be used for conventional VOR navigation when no offset waypoint is required.

Because the KNS-81 relies on radio-navigation data, correct frequency selection and suitable VOR/DME coverage are essential. Unlike modern GPS-based systems, it does not provide unrestricted point-to-point navigation and requires the crew to understand the relationship between stations, radials and distances.

We have recreated this system in the SK60 mod in a way that is user friendly and functional in the realm of DCS and lets the player fly IFR navigation including ILS approaches.

The system has three main parts – the **NAV panels**, the **HSI**, the **RMI**. In addition to them, the NAV system also uses the **EADI** for ILS guidance. These work together to give the pilot the information he needs to program and navigate along a desired route.

KNS-81 RNAV system



The KNS-81 RNAV system uses three units for programming and managing the navigation route. These are the **DME** unit (top), the **RNAV** unit (middle) and the **ADF** unit (bottom) as seen in the image above.

The **DME** unit shows distance information related to the current waypoint.

The **RNAV** unit is where the programming and managing happens. In the next section you're going to learn how to use it.

The **ADF** unit lets you enter a frequency to an **NDB** which then gives you a direction to the tuned station displayed on the **RMI** (the larger arrow) as seen in the image below. The other arrow is used by the RNAV unit and shows the direction to the currently selected waypoint.

Using the RNAV unit

The real KNS-81 RNAV unit has multiple modes. Our version currently only uses one of these – the RNV mode.

The unit has the following components:

1. Data screen with this info:
    a. Current mode and waypoint (WPT)
    b. Dialed frequency (FRQ)
    c. Selected radial (RAD)
    d. Selected ofset distance (DST)
2. Waypoint selector (left outer knob)
3. Mode selector knob (left inner knob – inop)
4. Data input encoders (right knobs – both inner and outer)
5. Control buttons
    a. USE – Confirm input data.
    b. RTN – Return to currently selected waypoint.
    c. RAD – INOP.
    d. CHK – a momentary spring loaded button that changes the information display so it shows current FROM radial (RAD) and distance (DST) from the tuned VOR.
    e. DATA – toggle between data input sections (FRQ, RAD, DST).

Programming the RNAV system

As mentioned in the navigation system overview, the KNS-81 can create offset waypoints. These waypoints are created by referencing a radial and distance from a VOR station. So in order to create a waypoint you need to:

1. Select waypoint number to be programmed.
2. Enter the frequency of the desired VOR station in the FRQ field.
3. Enter the radial (magnetic) from that VOR station to the waypoint.
4. Enter the distance in NM from that VOR station to the waypoint.

Once you're happy with the data you need to press USE in order to "load" that data into the selected waypoint. Once that is done you can continue to the next waypoint and repeat the process.

ILS

The navigation system has ILS guidance functionality. If you program a waypoint with a frequency that belongs to an ILS, the system will automatically recognize it as an ILS waypoint and provide you with glidepath and localizer guidance which is displayed on the EADI.

In order to activate the ILS overlay on the EADI you need to push the "Mode" button. This button will toggle between three modes: No ILS symbology, Glidepath and Localizer guidance bars, GP and LOC bars with deviation markers.

Following the navigation system

Once you're happy with your programmed route it's time to get airborne and fly it.

There are three main places where you'll find guidance in order to reach your desired destination. The first is one we mentioned earlier – the **DME** unit, which tells you the remaining distance in NM to the programmed waypoint. The DME unit can also give you closing speed in knots as well as a predicted time to point. This depends, however, on whether the system is able to calculate it or not.

The second unit that gives you information is the **RMI**, which we have also already mentioned. It shows the current heading you need to fly to reach the selected waypoint.

The third unit is the **HSI** which is a bit more special.

HSI navigation

The HSI has a compass rose that shows you your current magnetic heading (orange “needle” at the top). But it can do more than that. On the bottom part you have two knobs. The left knob moves the “**heading bug**” (orange “triangle”) which then follows the compass rose. Let’s say for example you know you want to fly on heading 265 after you’ve crossed a small lake. In that case you can turn the heading bug to that heading which makes it much easier for you to make the turn and establish yourself on that heading without needing to carefully read the compass rose.

The right knob is the **course selector** knob. It rotates the inner disk which has got a yellow “arrow” attached to it. This can be used to input a desired course you want to fly on to a programmed waypoint. Let’s say you have programmed waypoint 7 to be your target. You know you need to fly toward it on heading 140 in order to stay clear of terrain. You can then set your course selector knob so the arrow points to heading 140. The “bar” in the middle will then give you show you if you are on that course or if you’re deviating either to the left or to the right.

Saving & sharing a flightplan

You can have up to 10 programmed waypoints in the system. The programming is persistent, so the data you’ve entered into your system will be saved between sessions in a file called “Flightplan.lua” that’s placed in “Mods/aircraft/SK60B/Cockpit/Scripts/Systems”. If you’re flying with friends and want to share your flightplan, simply send that lua file to them and have them replace the lua file on their computer and their NAV system will be updated with the data you’ve programmed. This file can, of course, also be manually updated in a text editor (“Notepad++” for example) before you start DCS and it will then load the flightplan into your system when you start.

RADIOS

Overview

Our SK60 has two radios – the **FR31** (upper left corner) and the **FR33** (lower right corner).

The FR33 is the secondary radio and is fairly simple. It's a VHF (118.000 - 135.975MHz) radio which lets you tune a frequency with the three knobs on it. The frequency can be dialed in 25kHz steps.

FR31

The FR31 is the main UHF (223.000 - 407.976 MHz) radio of the SK60.Our version has two different modes:

- MHZ – Manual frequency input
- NR – Preset mode

MHZ

In MHZ mode you enter a desired frequency manually. You enter the first 5 digits of the desired frequency (it has 25kHz channel separation so the last 0 isn't needed). In order to enter a new frequency you first need to press the "CL" button (clear). That will empty the screen and let you input a new frequency. Once all five digits have been entered with a valid frequency you don't need to do anything else.

NR
The NR mode lets you select one of 10 pre-programmed stations. These are programmed in the mission editor and can be whatever frequency you want within the range of the FR31. In this mode, you won’t see the frequency on the screen. Instead you’ll see your selected preset which is represented as 100-110.

(BAS)
In the real aircraft there is a third mode – BAS which is also a preset mode tied to a more “rigid” database where each airfield, ATC sector, GCI and so on would have their pre-defined standard frequencies that wouldn’t change. This mode is currently not implemented in the SK60 mod.

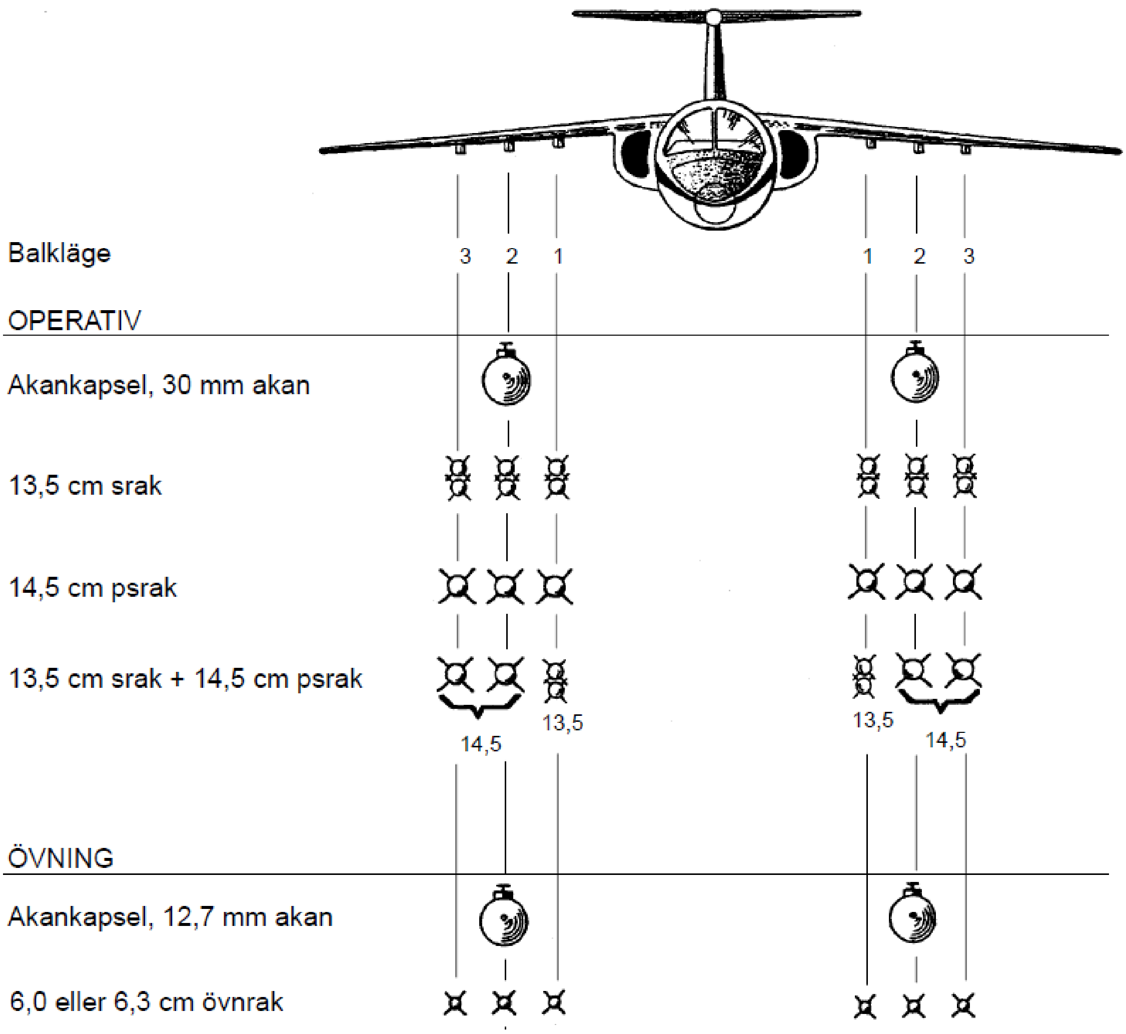
WEAPONS

Overview

The following weapons are available on the SK60 mod:

- **ARAK Rockets**
    - 13,5 cm HE rocket
    - 14,5 cm HEAT rocket
    - 6 cm practice rockets
    - 6,3 cm smoke marker rockets
- **AKAN 30mm gunpod**

They can be mounted according to this image:



Loadouts

Figure 1 - 12x HE rockets

Figure 2 6x HEAT rockets

Figure 3 - Mixed HE/HEAT rockets

Figure 4 – AKAN

Figure 5 - ÖRAK practice rockets

Figure 6 - Smoke marker rockets

Operating procedures

On the real SK60, the weapon settings are done prior to entering the aircraft on a box located behind the right seat. In the name of user-friendliness we've decided to take a slightly different approach.

Instead of the box we've got a menu system you can use to set up your weapons according to the mission you're gonna fly. The menu is activated via the "Configuration Menu Toggle" command in your keybinds. Via that menu you can then set the following (by using the "Configuration Menu Action" keybinding):

- **Select weapon**
  - *ARAK* or *AKAN*
- **Rocket firing mode**
  - *IMPULS* (One rocket at a time)
  - *SERIE* (Ripple as long as you hold the trigger)
- **Rockets per impulse**
  - *DUBBEL* (Two rockets per impulse)
  - *SINGEL* (One rocket per impulse)

When that is done you need to mount the gunsight. This is done via the keybind "Sikte montera". To remove it you use "Sikte parkera" instead.

Once your weapons are setup and the sight is mounted there are two more things you need to do:

- Set the master arm (BEVÄPNING) switch to TILL (on) which will supply power to your weapons and gunsight.
- Set open the safety lever/trigger on the stick (keybind).

Important note – the gunsight isn't very acurate so you're gonna need to practice to get it right.